

# CSG3101 APPLIED PROJECT

Intelligent Spatial Management System

# User Manual

25th January 2024

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# User Guide for Intelligent Spatial Management System

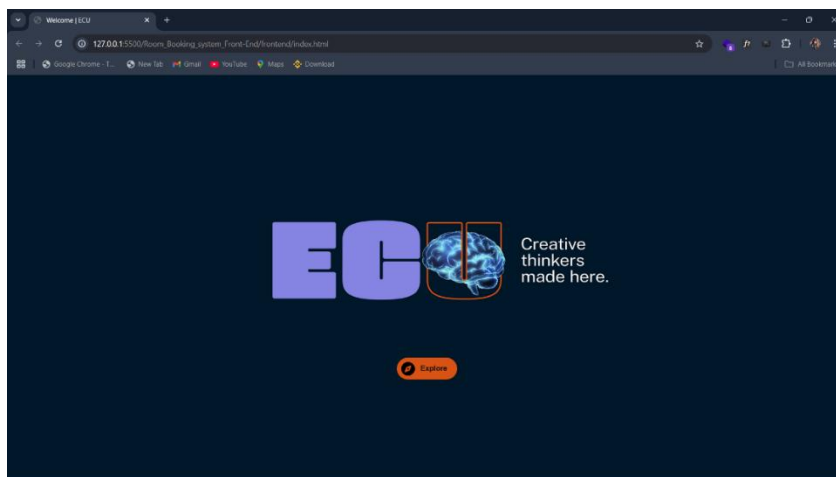
## Welcome to Intelligent Spatial Management System

Welcome to the Intelligent Spatial Management System, a web-based solution designed to efficiently manage library room bookings. This user manual will guide you through the website's features and functionalities, ensuring a seamless experience.

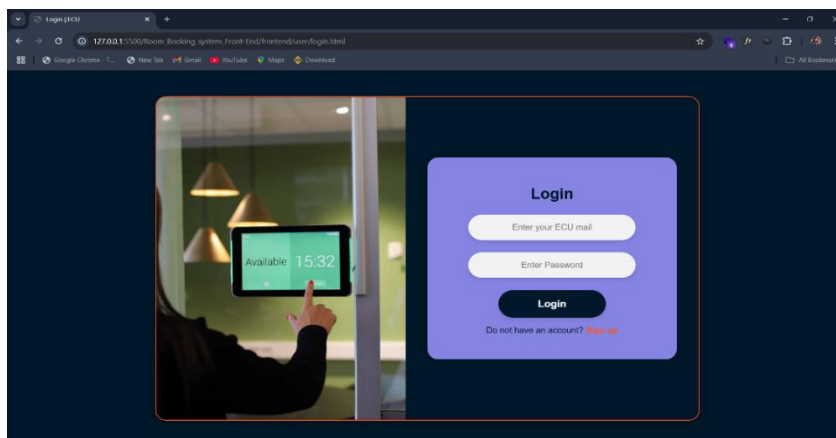
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## 1. Getting Started

### 1.1 Login and Sign Up

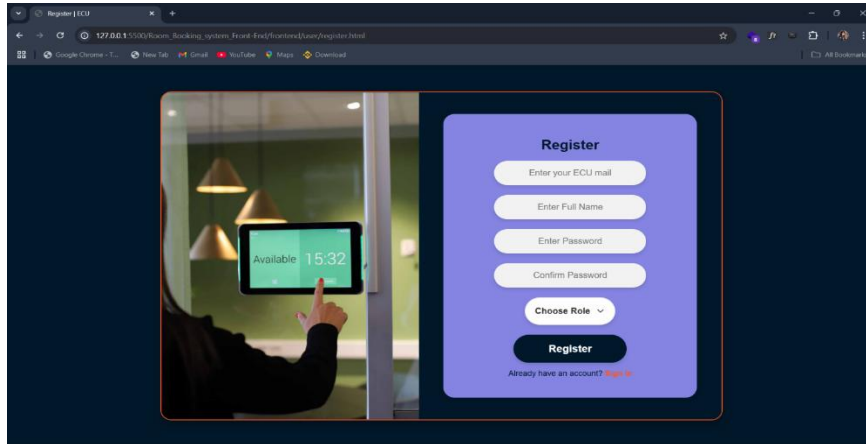


1. Press the **Explore** button on the homepage to start your journey
2. You will be directed to the login page.



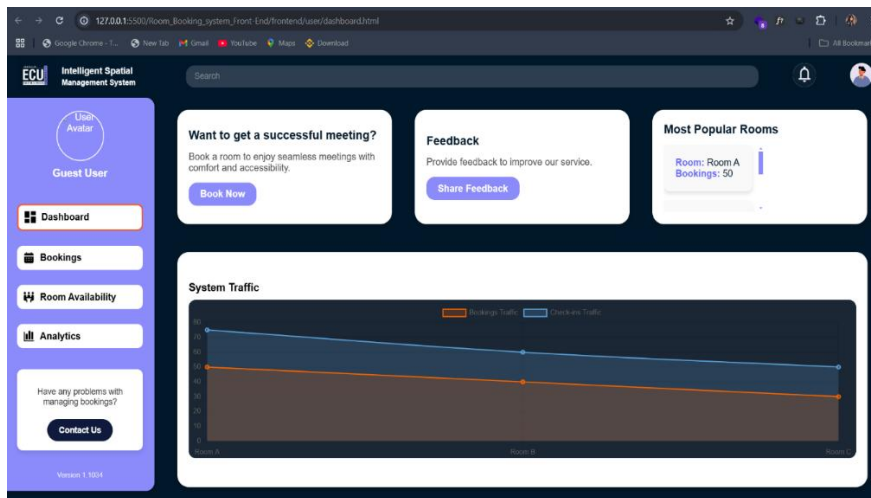
3.If you have an existing account, enter your credentials and click **Login** to access the dashboard.

4.If you are new, click **Sign Up**, fill in the registration form, and press **Register** to create an account.



### 1.2 Dashboard Navigation

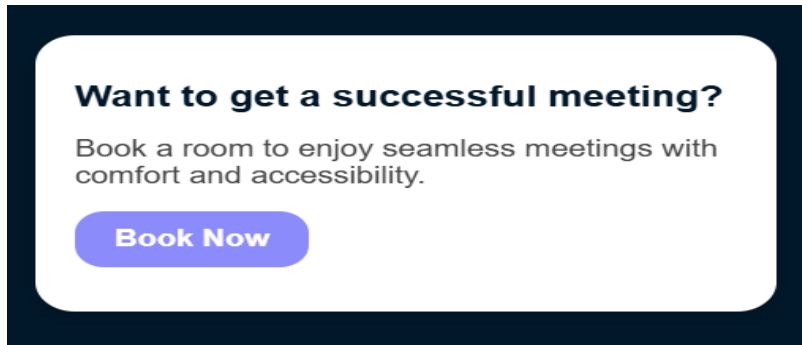
- If you are an **Admin**, you will be redirected to the admin dashboard.
- If you are a **User**, you will be directed to the user dashboard .



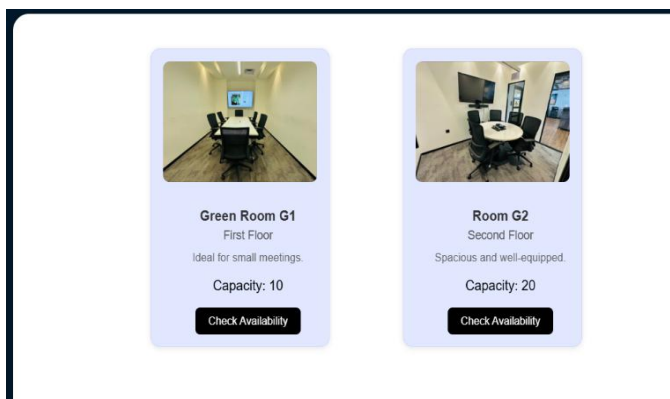
## 2. User Features

### 2.1 Book a Room

1. Click on the **Book Now** button on the dashboard or select **Room Availability** from the sidebar.

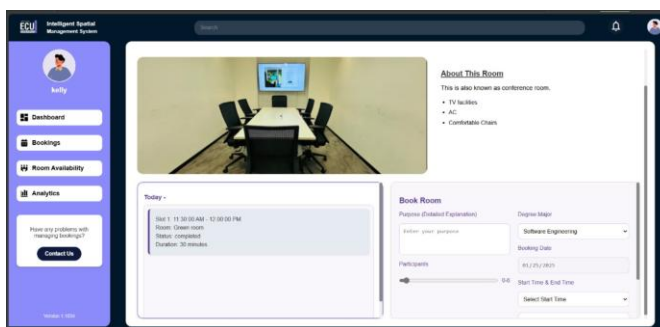


2. You will be redirected to the room selection page.

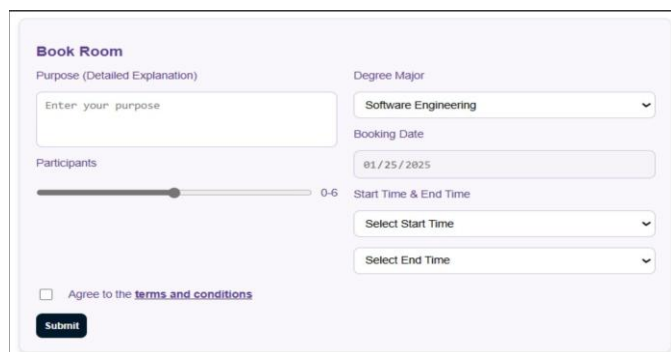


3. Choose a room and click **Check Availability** to display the booking form.

4. Then you will navigate to a UI that you can see room name, Room Amenities and its details and also Bookings for that Room and Room Booking Form.



5. Then If you want to Book this room Fill This form In here.



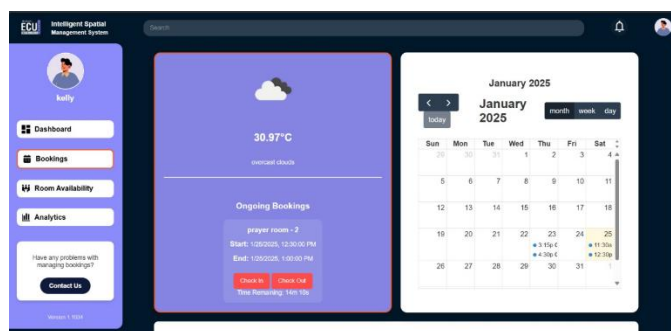
The 'Book Room' form is a light purple interface. It includes a 'Purpose (Detailed Explanation)' text area, a 'Degree Major' dropdown menu set to 'Software Engineering', a 'Booking Date' field set to '01/25/2025', and a 'Participants' slider set to '0-6'. There are 'Select Start Time' and 'Select End Time' dropdowns. At the bottom, there is an 'Agree to the terms and conditions' checkbox and a 'Submit' button.

6.Fill this Form and Then Agree to the Terms & Conditions and then Press **Submit Button**.

7.Then Your Booking will be placed and display in the right side Bookings.

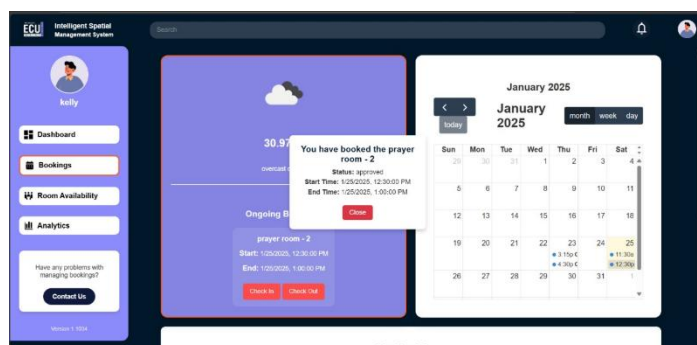
### 2.2 Check Bookings

1. Navigate to **Bookings** from the sidebar.



2.Your booked dates will be marked on the calendar, and your upcoming bookings are display in the left side.

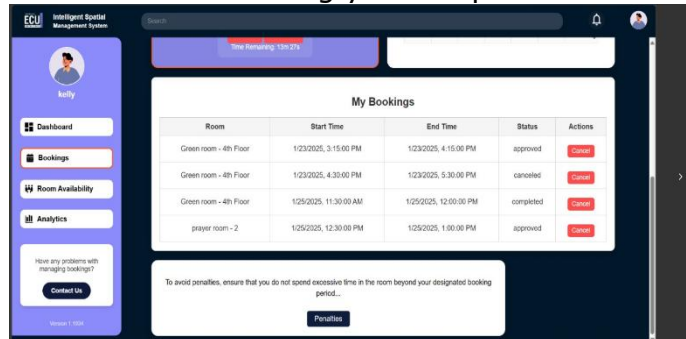
3.And You can Click on a Booking in the calendar and see the details of the bookings.



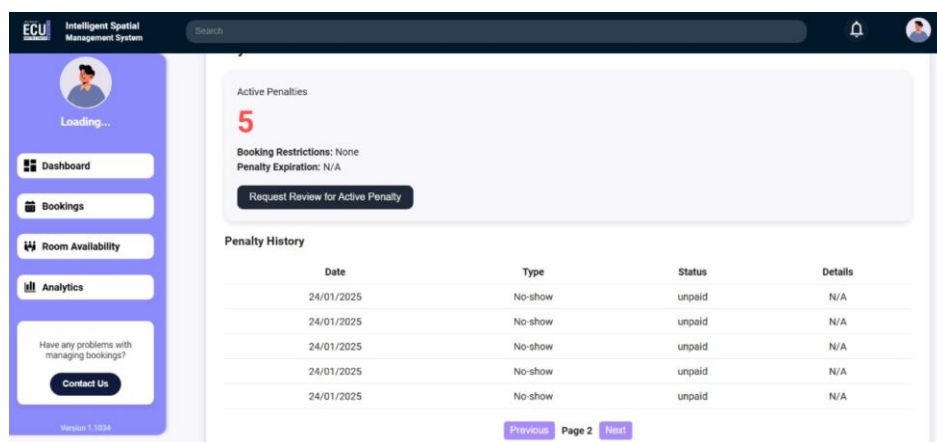
4.After Look Details you can press on **Close** button and close that pop up.

5.Once you come to the room you can Press **Check in** and once you go out of the room you can press **Check out** and your room booking will be over.

6. And in below your booking will be displayed and then from there if you want to cancel cooking you can press Cancel button.

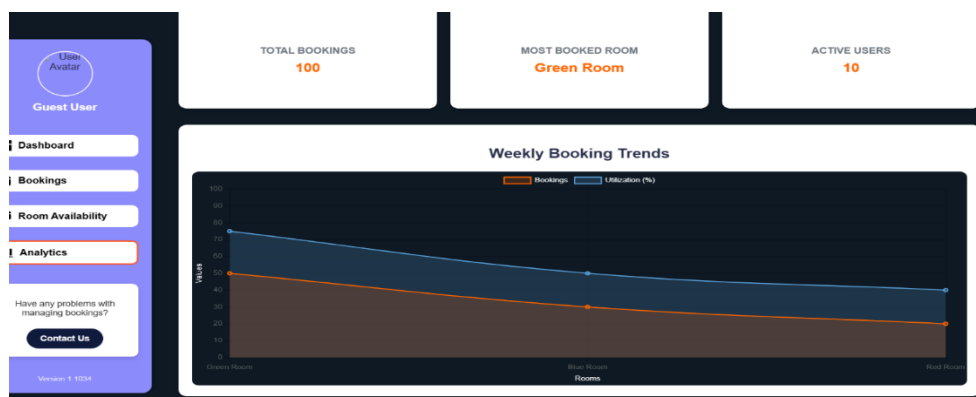


7. You can press on Penalties Button and see your Penalties.



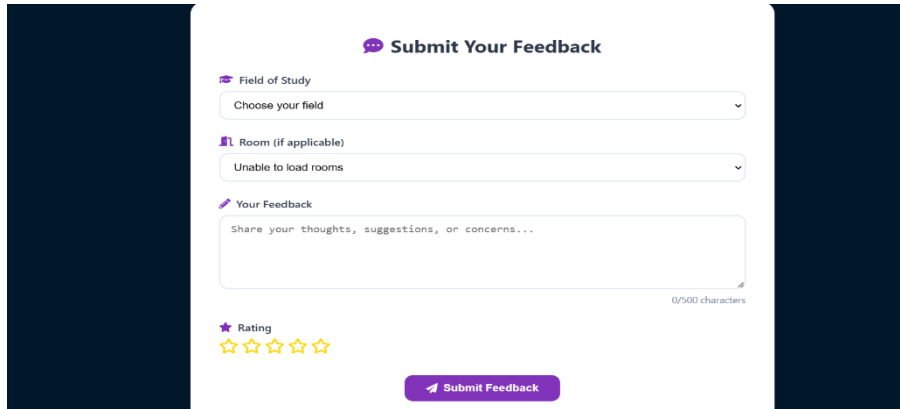
## 2.3 View Analytics

1. Click on **Analytics** in the sidebar to view booking trends and usage reports.



### 2.4 Contact Us & Feedback

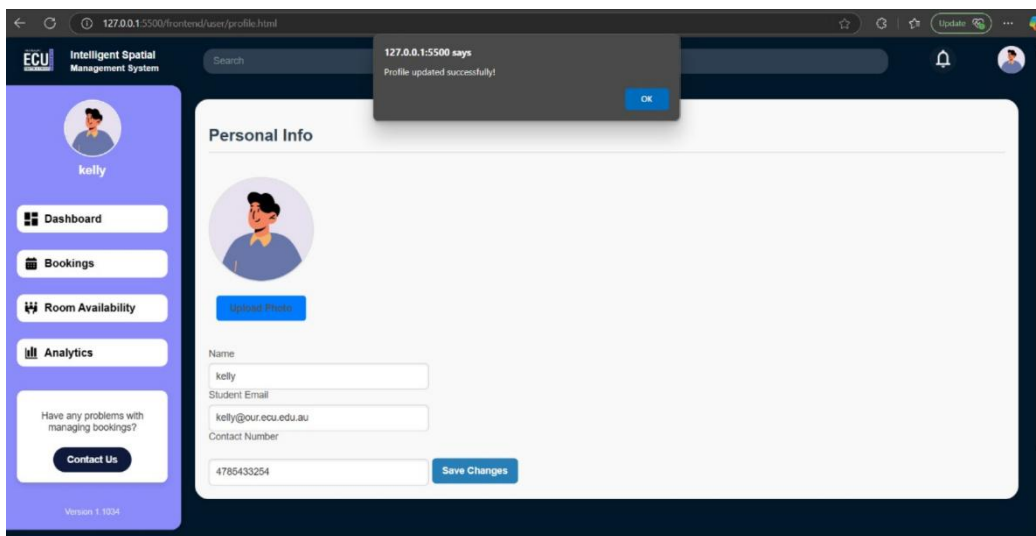
1. Select **Contact Us** from the sidebar.
2. Fill in the contact form and click **Submit** to send your queries.



The screenshot shows a 'Submit Your Feedback' form. It includes a dropdown menu for 'Field of Study' with the option 'Choose your field', another dropdown for 'Room (if applicable)' with the option 'Unable to load rooms', a text area for 'Your Feedback' with the placeholder 'Share your thoughts, suggestions, or concerns...' and a character count '0/500 characters', and a star rating system with five stars. A 'Submit Feedback' button is at the bottom.

### 2.5 Personal Info

1. Click the **User Icon** in the header and select **Profile Info** to view your details.
2. In there you can change the phone number but you do not have access to change your student email and the name.

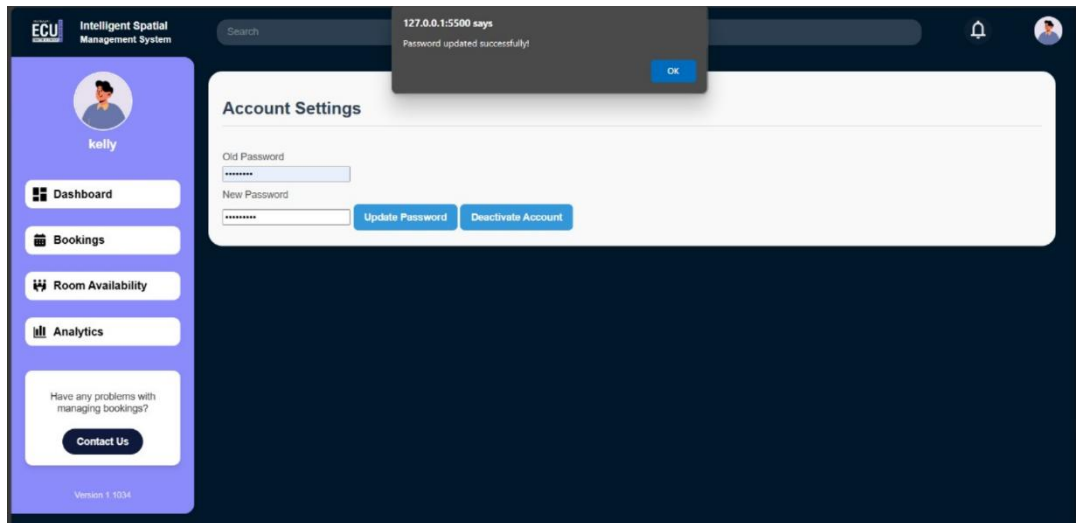


The screenshot shows the 'Personal Info' profile page. A notification banner at the top says '127.0.0.1:5500 kays Profile updated successfully!'. The page has a sidebar with navigation links: Dashboard, Bookings, Room Availability, Analytics, and a 'Contact Us' button. The main content area shows a profile card with a user icon and an 'Upload Photo' button. Below the card are input fields for Name (filled with 'kelly'), Student Email (filled with 'kelly@our.ecu.edu.au'), and Contact Number (filled with '4786433254'). A 'Save Changes' button is at the bottom right.



### 2.7 Account Settings

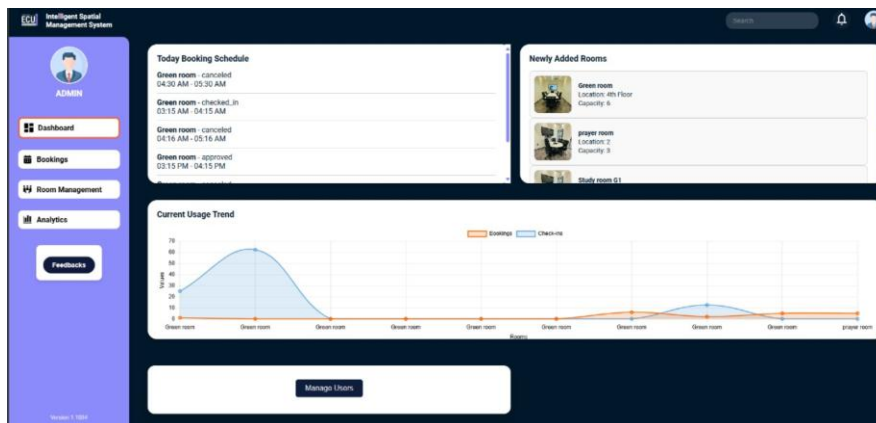
1. Click the **User Icon**, go to **Account Settings**, update your details, and press **Save**.



2. You can Change Your Password here.
3. You have to First put your **Old password**(current password).
4. Then Put the new password you want and then press **Update password**.
5. If you want to deactivate your account just simply press **Deactivate Account** Button.

### 3. Admin Features

If you are an admin You will navigate to a different dashboard.



1. You can View all the bookings for today.
2. And also, You can See Recently added Rooms.
3. And you can see Analytics graphs for current usage trends.
4. And from the below **Manage Users** Button You can Manage users.

### 3.1 Room Management

#### Update Room Info

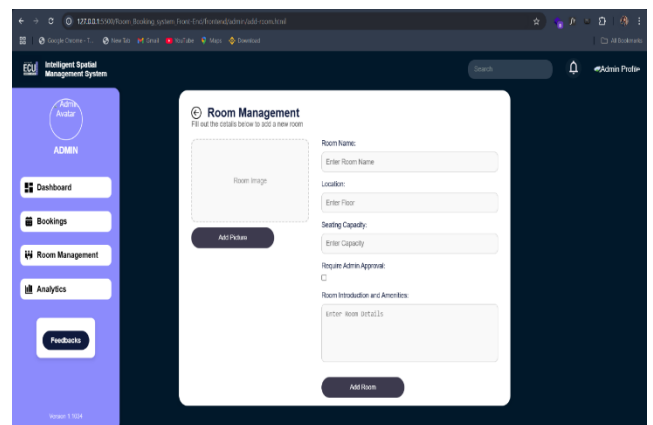
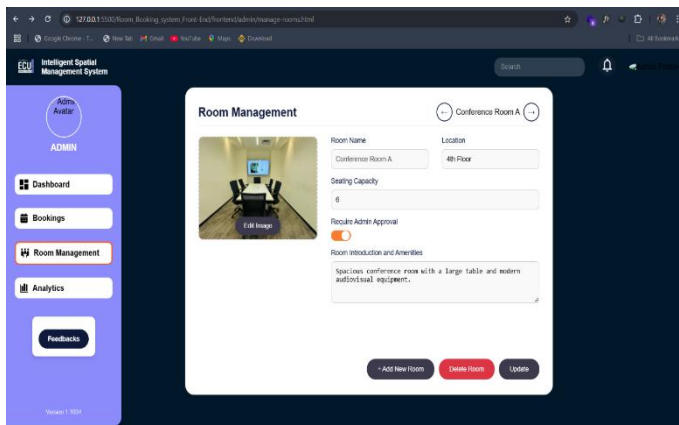
1. Navigate to **Room Management** from the sidebar.
2. Update the required details and press **Update**.
3. Use navigation keys to browse through rooms.

#### Delete a Room

1. Click on the **Delete** button to remove a room permanently from the system.

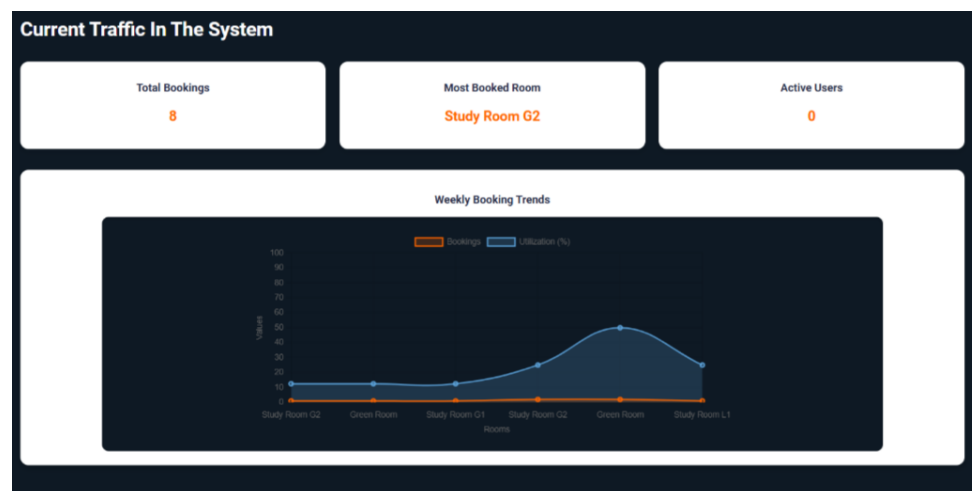
#### Add a Room

1. Click **Add New Room** and enter the required information.
2. Press **Add Room** to save the new room.



### 3.2 View Analytics

1. Click **Analytics** in the sidebar to access reports on room usage and bookings.



### 3.3 View Bookings

1. Navigate to **Bookings** to view all reservations by date.

The screenshot shows the "Bookings" page of the ECU Intelligent Spatial Management System. It includes a sidebar with navigation options (Dashboard, Bookings, Room Management, Analytics, Feedbacks) and a main content area with filters and a table of bookings.

**Filters:** All, Search by User, Search by Room, Apply Filters, Add Booking

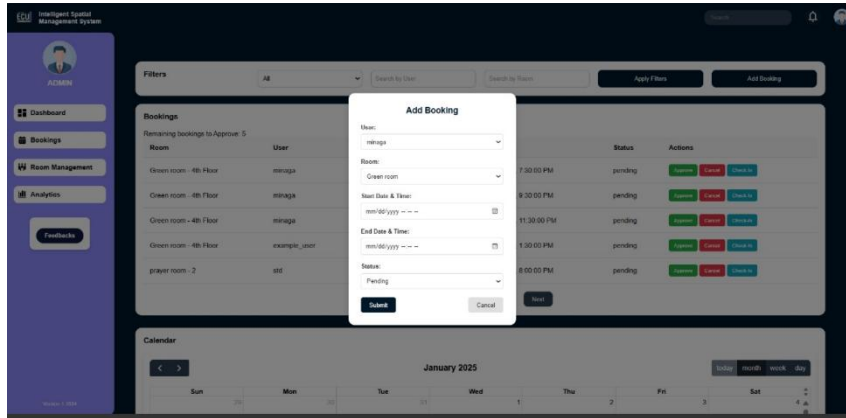
**Bookings:** Remaining bookings to Approve: 6

| Room                   | User   | Start Time             | End Time               | Status     | Actions                 |
|------------------------|--------|------------------------|------------------------|------------|-------------------------|
| prayer room - 2        | kelly  | 1/25/2025, 12:30:00 PM | 1/25/2025, 1:00:00 PM  | pending    | Approve Cancel Check-In |
| Green room - 4th Floor | std    | 1/24/2025, 7:30:00 PM  | 1/24/2025, 8:00:00 PM  | completed  | Approve Cancel Check-In |
| Green room - 4th Floor | kelly  | 1/25/2025, 11:30:00 AM | 1/25/2025, 12:00:00 PM | completed  | Approve Cancel Check-In |
| Green room - 4th Floor | minaga | 1/23/2025, 3:15:00 AM  | 1/23/2025, 4:15:00 AM  | checked_in | Approve Cancel Check-In |
| Green room - 4th Floor | minaga | 1/23/2025, 4:16:00 AM  | 1/23/2025, 5:16:00 AM  | cancelled  | Approve Cancel Check-In |

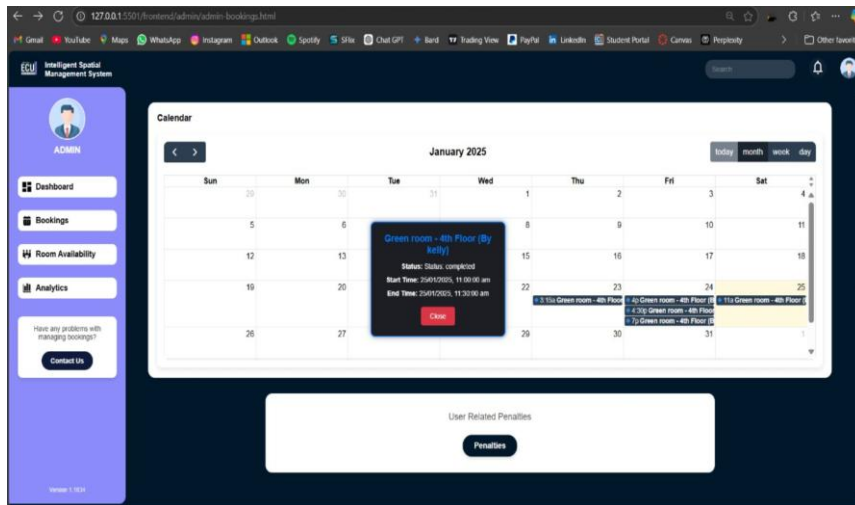
Previous Next

2. You can See all The Booking made By users You can approve them by clicking **Approve** Button
3. You can Cancel Bookings by simply clicking **Cancel** Button.
4. If the users come to the room and booking is going on Admin can Press **Check In** Button that means booking completed successfully.
5. From here if you want to Add a manual booking you can Click **Add Booking**.

## User Guide



6.Fill this form and then **Submit** Then Your booking will be added.



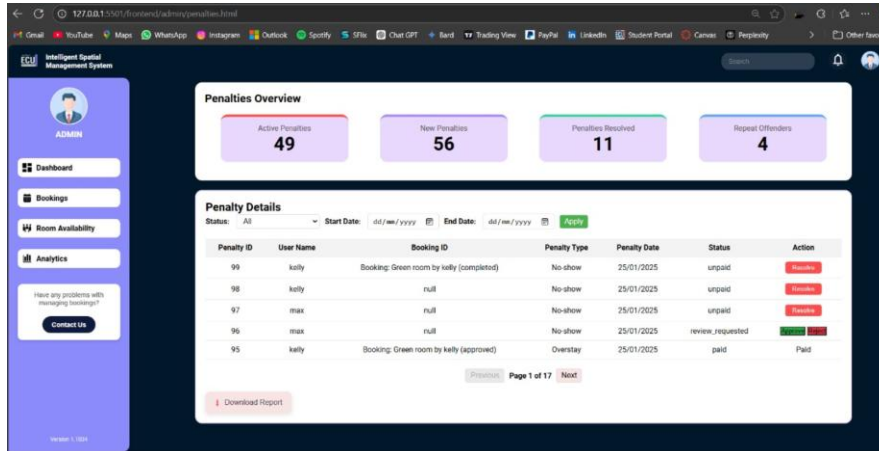
7.Below That In the Calendar all the bookings display in the calendar.

8.From there you can click on one booking and above pop up will appear by with the room name, floor room, and status, start time and the end time.

9. And to get out of this pop up click on **Close** button.

### 3.5 View Penalties

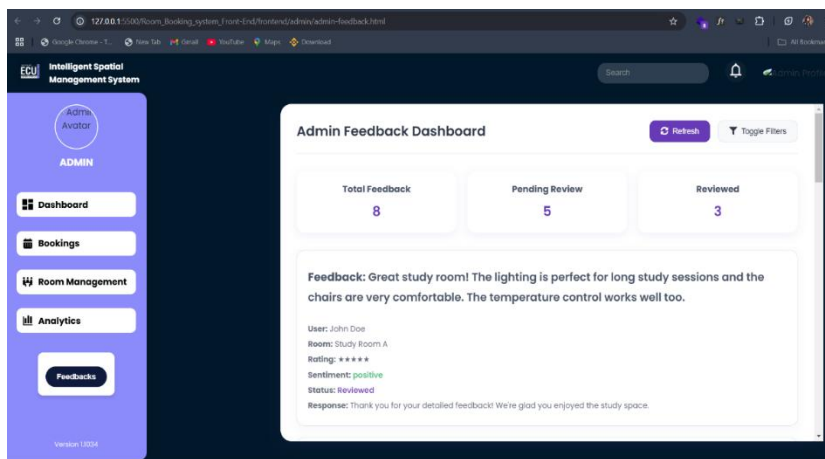
Click on the Penalties Button and then You will Navigate to penalties section.



1. Navigate to **Penalties Overview** to view all penalty details.
2. Use filters to refine results and press **Apply**.
3. Click **Download Report** to save penalty data for analysis.
4. You can click on **Approve** button to approve a penalty.
5. And if you want to Resolve Penalty by the beginning you can click on **Reject** Button.
6. And if you want to resolve penalty that means penalty is over you can click on **Resolve** button.

### 3.4 Read Feedback

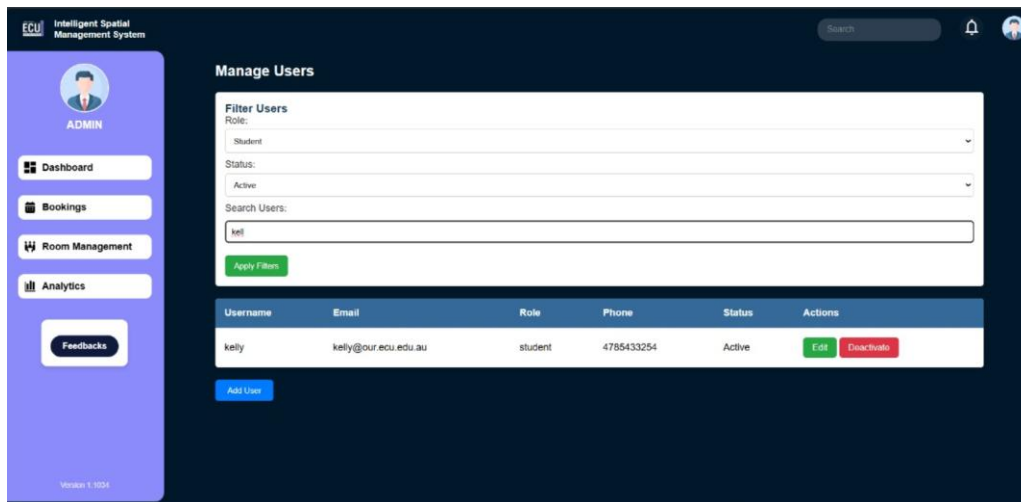
1. Click **Feedback** in the sidebar to view, filter, and respond to user feedback.



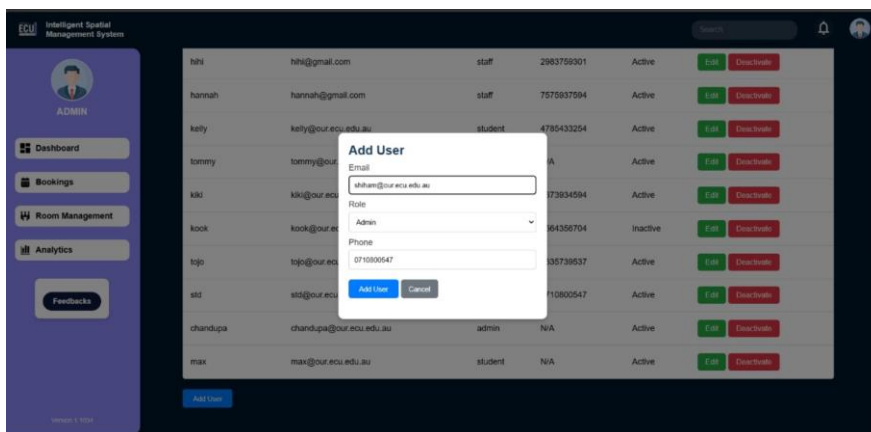
2. You can see overview of feedbacks at the top And you can refresh the feedbacks and also You can apply a filter.
3. To apply a filter you can click on **Toggle Filters** and then apply filters as you need then press **Apply Filter** Button then feedbacks will filter according to your desire.

### 3.5 User Management

1. Click on **User Management** Button in the Dashboard in there You can access User Management.



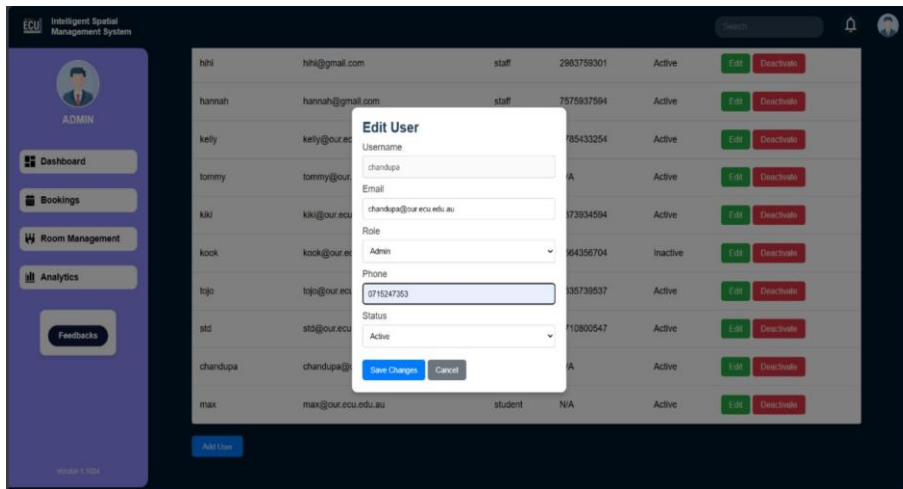
2. This will display all the users and you can Use a filter to filter out the desired users you want.
3. What you have to do is to Fill those filters above and then press **Apply Filters** button then it will filter out the users you want.
4. And admin has access to manually add a user what you have to do is to press on **Add User** button.



5. There You can fill User Email, Role, and then Phone number and Then Press **Add User** button then user will be added to system if do not want press **Cancel** Button.

**\*\*Important: You do not need to create the password because password will be automatically generated.\*\***

6. You can Click on the **Edit** button and then admin can Change user details.



7. You can add details in the relevant field you want and then press **Save Changes** If you want to keep the previous data click on **Cancel** Button.

8. And at the same time admin has access to remove and user just simply click on **Deactivate** button in User Management Dashboard.

## 4.ChatBot

### Overview

The chatbot is an AI-powered online assistant integrated into the Room Booking System (RBS) to help users navigate features, check room availability, make bookings, and manage existing reservations. It enhances user experience by providing quick and accurate responses.

### Features

#### 1. Room Availability Inquiry

- Users can ask the chatbot to check available rooms based on criteria like location, capacity, or specific room names.
- **Example:** "Which rooms are free right now?"

#### 2. Room Booking

- The chatbot can assist with booking a room by specifying the room name, date, time, purpose, and capacity.
- **Example:** "Book Green Room from 2 PM to 4 PM for 5 people for a team meeting."

### 3. Cancel Bookings

- Cancel an existing booking directly through the chatbot.
- **Example:** "Cancel my booking for Green Room tomorrow."

### 4. Identify Room Details

- Query details about specific rooms, such as their location and capacity.
- **Example:** "Tell me about the Green Room."

### 5. Search Rooms

- Search for rooms matching specific criteria, such as location or minimum capacity.
- **Example:** "Find a room with at least 10 seats on the 4th floor."

### 6. General Assistance

- The chatbot can answer basic questions about its functionality or provide guidance.
- **Example:** "How do I book a room?"

## How to Access the Chatbot

### 1. Login Requirement

- Users must log in to interact with the chatbot.
- Ensure you are logged into your account to access personalized features like bookings and cancellations.

### 2. Chatbot Interface

- Located at the bottom-right corner of the web application or available on the chatbot page.
- Click the chatbot icon to open the chat window.

### 3. Session-Based Interaction



- The chatbot remembers your session, so you don't need to repeat basic details like your name or role during a session.

## Using the Chatbot

### 1. Initiating a Conversation

- Open the chatbot and type your query into the text box.
- Press Enter or click the send button to receive a response.

### 2. Asking Questions

- Keep your questions clear and specific for the best results.
- **Example:**
  - "Is Green Room available from 10 AM to 12 PM?"
  - "Cancel my booking for tomorrow."

### 3. Booking a Room

- Specify the room name, time, capacity, and purpose.
- **Example:** "Book Study Room G1 from 3 PM to 4 PM for 3 people for a study session."

### 4. Canceling a Booking

- Provide details of the booking to cancel.
- **Example:** "Cancel my meeting booking for the Green Room tomorrow."

### 5. Checking Availability

- Ask about free rooms or specify criteria like location or capacity.
- **Example:**
  - "What rooms are free right now?"
  - "Find me a room on the ground floor for 4 people."

### 6. Room Details

- Ask the chatbot for details about a specific room.
- **Example:** "What is the capacity of the Study Room L1?"

### Best Practices

- **Be Specific:** Provide detailed information about your request for better results.
- **Use Natural Language:** You can communicate naturally with the chatbot.
- **Avoid Abbreviations:** Write full names and times to avoid misunderstandings.

### Troubleshooting

#### 1. The Chatbot is Not Responding

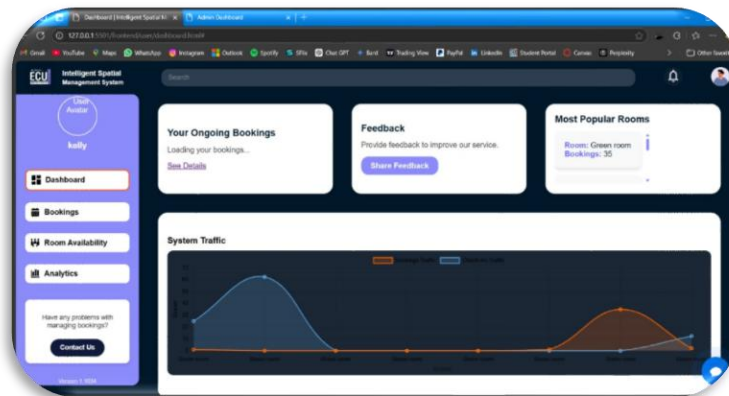
- Ensure you are logged in.
- Check your internet connection.
- Refresh the page and try again.

#### 2. Incorrect Response

- Ensure your question is clear and complete.
- If the issue persists, contact support.

#### 3. Session Issues

- If the chatbot doesn't recognize your session, log out and log in again.



## User Guide



**We hope this user manual helps you navigate the Intelligent Spatial Management System with ease. Happy exploring!**